MULT20015 Elements of Quantum Computing

Practice Class 6

Welcome to Practice Class 6 of MULT20015 Elements of Quantum Computing, covering exercises relevant to the material presented in lectures in Week 6.

The purpose of this week's exercises is to:

- Implement RSA with small parameters
- Investigate the quantum phase estimation algorithm

1 RSA parameters for small inputs

In this exercise we will work through some RSA key generation examples with small parameters. Please go through the RSA encryption and decryption function definitions. Your task is to determine RSA's public and private key pairs given the two primes required. You will make use of extended Euclid's algorithm finding greatest common divisor between two numbers in the process.

Given N = PQ; where P, Q are primes, and the public and private key pair (e, d) is computed such that

$$ed \equiv 1 \bmod \phi(N), \text{ where } \phi \text{ is Euler's totient function.}$$

To find d given e, we will use the extended Euclid's algorithm: For any integers a and b, there exist integers x and y such that:

$$\gcd[a; b] := ax + by.$$

Given e and $\phi(N)$, then run the extended Euclid's algorithm with input (e, $\phi(N)$), then you get $\gcd[e, \phi(N)] = e\,x + \phi(N)\,y$, which has to be equal to 1, since e is chosen such that it is relatively prime to $\phi(N)$. Taking (mod $\phi(N)$) on both the sides of the equation gives, $e\,x = 1 \pmod{\phi(N)}$. Hence x will be inverse of e modulo N, so d = x.

You can compute the steps of the extended Euclid's algorithm manually following the algorithm given below:

GCD(a,b) $a = q_1 b + r_1$ $b = q_2 r_1 + r_2$ $r_1 = q_3 r_2 + r_3$ $r_2 = q_4 r_3 + r_4$ … (1) $r_{n-2} = q_n r_{n-1} + r_n$ (2) $r_{n-1} = q_{n+1} r_n + r_{n+1}$, **where $r_{n+1} = 1$ (GCD exists)** (3) $r_n = q_{n+2} r_{n+1} + r_{n+2}$, **where $r_{n+2} = 0$**	Now we can perform a back substitution to get d as follows: From (2) we get $r_{n+1} = 1 = r_{n-1} - q_{n+1} r_n$ We know r_n from (1), so we can substitute $= r_{n-2} - q_{n+1}(r_{n-2} - q_n r_{n-1})$ We continue this for each r while simplifying each step until we can represent the r_{n+1} in terms of b.

Exercise 1.1 Given the p=13 and q=7, compute the RSA parameters n and $\phi(n)$. For the RSA algorithm to work, it requires two coefficients – e and d. Where e represents the encryption component (generally the public key) and d represents the decryption component (generally the private key).

Suppose $\phi(n) = 72$, using the extended Euclid's algorithm given above, calculate the value of d such that $ed = 1 \bmod \phi(n)$ for

a) e=5
b) b=7

You can use the online Magma calculator to run the extended Euclid's algorithm. Magma computation is available online at

http://magma.maths.usyd.edu.au/calc/

Magma Calculator

Magma Calculator Enter your code in the box below. Click on "Submit" to have it evaluated by Magma. Calculations are restricted to 120 seconds. Input is limited to 50000 bytes. Running Magma V2.25-4. Magma is maintained and distributed by the Computational Algebra Group, School of Mathematics and Statistics, University of Sydney.
magma.maths.usyd.edu.au

For example:

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>XGCD(13,25);
>1 2 -1
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Indicating that gcd = 1 = 13*2 + (-1) * 25;

ExtendedGreatestCommonDivisor(m, n) : RngIntElt, RngIntElt -> RngIntElt,

RngIntElt, RngIntElt
Xgcd(m, n) : RngIntElt, RngIntElt -> RngIntElt, RngIntElt, RngIntElt
XGCD(m, n) : RngIntElt, RngIntElt -> RngIntElt, RngIntElt, RngIntElt

The extended GCD of m and n; returns integers g, x and y such that g is the greatest common divisor of the integers m and n, and g = x.m + y.n. If m and n are both zero, g is zero; otherwise g is always positive. If m and n are both non-zero, the multipliers x and y are unique.

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Exercise 1.2: Given the encryption and decryption formulas for RSA as follow: $C = M^e \ mod\ n$; $M = C^{\ d}\ mod\ n = (M^e)^d\ mod\ n = M\ mod\ n$. Perform encryption and decryption for the given values of p, q, e and M. You can use the online Magma calculator.

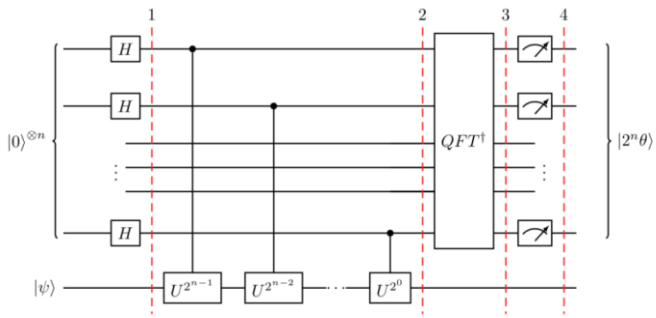
a. p =3, q =13, e =5; M =10.
b. p = 5, q =11, e =13; M =12.
c. p = 11, q =7, e = 7; M = 7.
d. p = 31, q=43, e =23, M = 11.
e. p = 54979, q = 59981, e = 1193, M = 101.

2 Quantum Phase Estimation

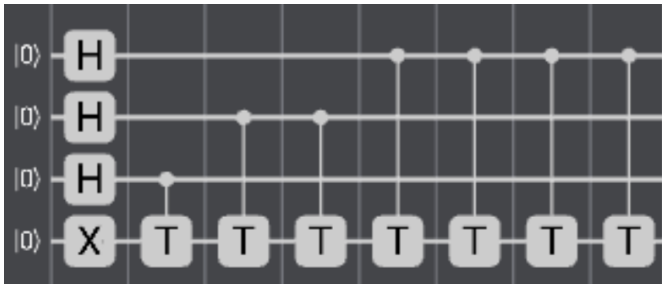
In this exercise we will construct the Quantum Phase Estimation (QPE) circuit described in the lectures, to determine the eigenvalue of the T gate, corresponding to the $|1\rangle$ state. Recall that if

$$U|\psi\rangle = \exp(i2\pi\theta)\,|\psi\rangle,$$

then QPE allows us to estimate θ using a quantum computer. In this section we will use QUI to construct the QPE circuit with U=T and $|\psi\rangle = |1\rangle$.



Exercise 2.1 First let us initialize the circuit. Construct the following circuit to initialize the state $|\psi\rangle$, and apply the controlled-U gates:

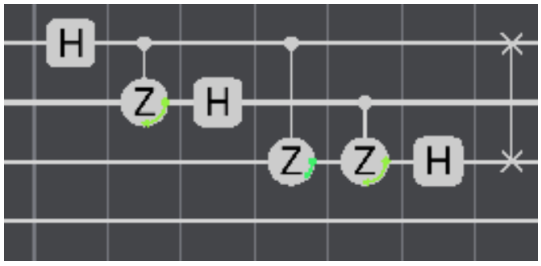


Note that there are 1, 2, and 4 controlled-T gates controlled from the 1s, 2s and 4s binary digits of the upper register. Also, note that the bottom register is initialized (by applying an x-gate) to the eigenstate.

Exercise 2.2 Having made the circuit above, examine the phase of each of the states. Fill in the following table showing each populated state and the corresponding phase:

State	Phase (Radians)
$\|0001\rangle$	0

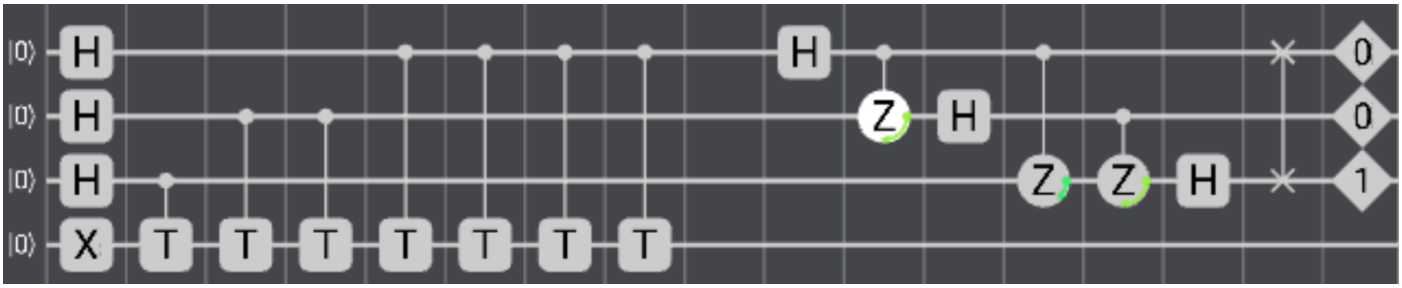
Exercise 2.3 After your existing circuit construct the Quantum Fourier Transform on the three qubits in the upper register.



The angles of the controlled-Z rotations shown are:

(a) $-\pi/2$ (with a global phase of $-\pi/4$)
(b) $-\pi/4$ (with a global phase of $-\pi/8$)
(c) $-\pi/2$ (with a global phase of $-\pi/4$)

Exercise 2.4 Finally, measure the three qubits in the upper register. Your circuit should now look like this:



The measured value relates to the original eigenvalue by

$$\theta = \frac{y}{2^n}$$

where y is the measured value, n is the number of qubits in the upper register and θ is the angle we would like to find. Verify that your circuit finds:

$$\theta = \frac{1}{8}$$

corresponding to an eigenvalue of the T operator of $\exp(i2\pi\theta) = \exp(i\pi/4)$ and the eigenstate, which we prepared in the bottom register, of $|1\rangle$.

Exercise 2.5 Repeat the Quantum Phase Estimation algorithm (above) to find the eigenvalue of the S gate, corresponding to the eigenstate $|1\rangle$.

Exercise 2.6 Repeat the Quantum Phase Estimation algorithm (above) to find the eigenvalue of the Z gate, corresponding to the eigenstate $|1\rangle$.